PYTHON LAB ASSIGNMENT 9

BUSINESS THESIS SUBMISSION – PROJECT REVIEW 2

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23MIC0020

LogScope Business Thesis Audio:

https://drive.google.com/file/d/1eIwubqMVQA8M6GdsszXqrdCntXqyXUHL/view?usp=drive_link

Transcript

00:00 Good evening, I'm P. Ganesh Prakadesh and my project is called Logscope. So the problem I'm trying to solve is that when software applications encounter errors, developers will have to manually go through a lot of logs and each logs will have hundreds or thousands of lines.

00:17 And they'll have to go through everything manually to understand what went wrong and this takes a lot of time and makes troubleshooting very difficult. So Logscope makes this process simpler. Users can just upload various format of logs like SQL, Python, Java and even Minecraft logs. The website will analyze them, identify the errors, group repeated issues into one and provide possible causes with solutions.

00:45 It also displays the statistics and visual summaries of the analysis made through a dashboard. So the main goal would be to reduce manual effort and help users understand problems faster. So the main users of Logscope are software developers, database administrators, server owners, IT support teams, and even students.

01:07 For example, a Java developer might upload application logs to investigate a database error. Similarly, Python developers can use it to investigate exceptions, while database users can use it to analyze SQL related errors.

01:21 Instead of manually searching through large files, users get an organized view of important errors, their severity, and possible fixes. And this can reduce troubleshooting effort, improve productivity, and help teams make better use of their time. Especially for students who are just learning about these errors, Logscope will be really helpful for them.

01:44 Now the frontend uses HTML, CSS and JavaScript to build the website and display the results. But the backend uses Python with Flask. Flask handles all the file uploads, API communication and request endpoints.

01:59 Between the website and the analysis engine. The analysis engine basically parses the logs uploaded, it'll classify the errors and it will group repeated issues. And it uses the predefined rules that are already in the database to suggest possible causes and fixes. And for the storage, we use PostgreSQL through Neon, which is a cloud-based database. It stores reusable knowledge.

02:27 Which will be helpful to find relevant solutions to be retrieved when similar errors appear again. These uploaded logs are intended to be processed temporarily rather than permanently. And to conclude, Logscope basically aims to make log analysis simpler, save the troubleshooting times, help

the user identify and understand errors more efficiently so that they can spend time on fixing them rather than trying to identify and understanding them.

02:56 Thank you.